Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - IV

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1. Creating a NumPy Array

A NumPy array is created using the array() function.

```
import numpy as np
```

```
a = np.array([1,2,3])
```

```
print(a)
```

```
[1 2 3]
```

2. Basic Array Operations

NumPy supports element-wise arithmetic operations such as addition, subtraction, multiplication, and division.

```
a = np.array([1,2,3])
print(a+10)
print(a*2)
```

```
[11 12 13]
[2 4 6]
```

3. Array Slicing

Slicing is used to access a specific range of elements from an array.

```
a = np.array([1,2,3,4,5,6,7,8,9,10])
a[0:5]
```

```
array([1, 2, 3, 4, 5])
```

4. Sorting an Array

The sort() function arranges array elements in ascending or alphabetical order.

```
a = (["Divyanshu","Manthan","Yash","Allwin"])
print (np.sort(a))
```

```
['Allwin' 'Divyanshu' 'Manthan' 'Yash']
```

5. Filtering an Array

Filtering returns elements that satisfy a given condition.

```
a = np.array([10,20,300,400])
a[a > 50]
```

```
array([300, 400])
```